Jagan K K

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Summary

Computer Science undergraduate (B.Tech, May 2026) experienced in full-stack web development, AI/ML, and AR game development using React, TypeScript, Node.js, Python, and LangChain. Built RAG pipelines (ChromaDB, Gemini), deployed apps on Vercel, and implemented real-time computer vision (OpenCV, YOLOv8, MediaPipe). Seeking a software engineering role to contribute and grow.

Technical Skills

Languages: Python, TypeScript, JavaScript, C, Java, SQL, HTML, CSS
Frameworks: React, Node.js, TanStack Start/Router, Vite, Tailwind CSS, FastAPI, Flask, LangChain, LangGraph, Vercel AI SDK
AI/ML: RAG, ChromaDB, Google Gemini, Groq, Tavily, OpenRouter, OpenCV, YOLOv8, MediaPipe
Databases & Tools: MongoDB, PostgreSQL, Supabase, MySQL, Prisma, Git, Docker, Vercel, Bun, Unity, AR Foundation

Education

St. Thomas College of Engineering and Technology, Kannur, Kerala — B.Tech in Computer Science (Sep 2022 – May 2026 Expected). CGPA: 8.38/10.

Indira Gandhi Public School, Kannur, Kerala — Higher Secondary Education (2019 – 2021). Plus Two: 79%, SSLC: 78%.

Internship

DevOps Intern — Kerala Vision Broadband Ltd. Designed and developed a secure, automated system for backing up network device configurations, with role-based access control and scheduled backups. Implemented a configuration comparator, device-specific command mapping, and WhatsApp notifications on backup status.

Projects

- **Cerbi: Retro OS Desktop Experience** (TypeScript, React, TanStack Start, Supabase, Tailwind CSS v4) — Built a full-stack app recreating a desktop OS as a retro game-style experience (draggable windows, pixel-art UI) with real-time code chat via Supabase Realtime and AI-generated news summaries via OpenRouter. Deployed on Vercel.
- **Zithr: Live Streamer & Browser Caching** (Python, FastAPI, MongoDB, React, Docker) — Built a music streaming platform importing Spotify playlists via CSV, resolving tracks to free streams, and caching audio in-browser via IndexedDB; exposed a key-protected REST API (FastAPI + MongoDB) and containerized with Docker.
- **Stateful RAG Pipeline with Vector Deduplication** (Python, LangChain, Google Gemini, ChromaDB) — Engineered a RAG system (LangChain, Gemini embeddings, ChromaDB) with stateful incremental indexing via sha256 hashing (SQLRecordManager) to skip unchanged PDF chunks, eliminating duplicate embedding API costs.
- **AI AR Data Augmentation** (Python, Flask, Unity, OpenCV, MongoDB) — Developed an admin web app and user mobile app delivering documents without login, gated by geofencing and YOLO-based object detection. Applied AI depth estimation and document summarization to turn PDFs into interactive 3D AR cards (Unity/AR Foundation).
- **SketchPlay AI** (Python, OpenCV, YOLOv8, MediaPipe) — Collaborated on a real-time AR game where hand-drawn sketches become 2D game platforms, using trained YOLOv8 and RNN models; optimized to 30 FPS with MediaPipe hand-gesture controls.

Languages

English, Hindi, Malayalam